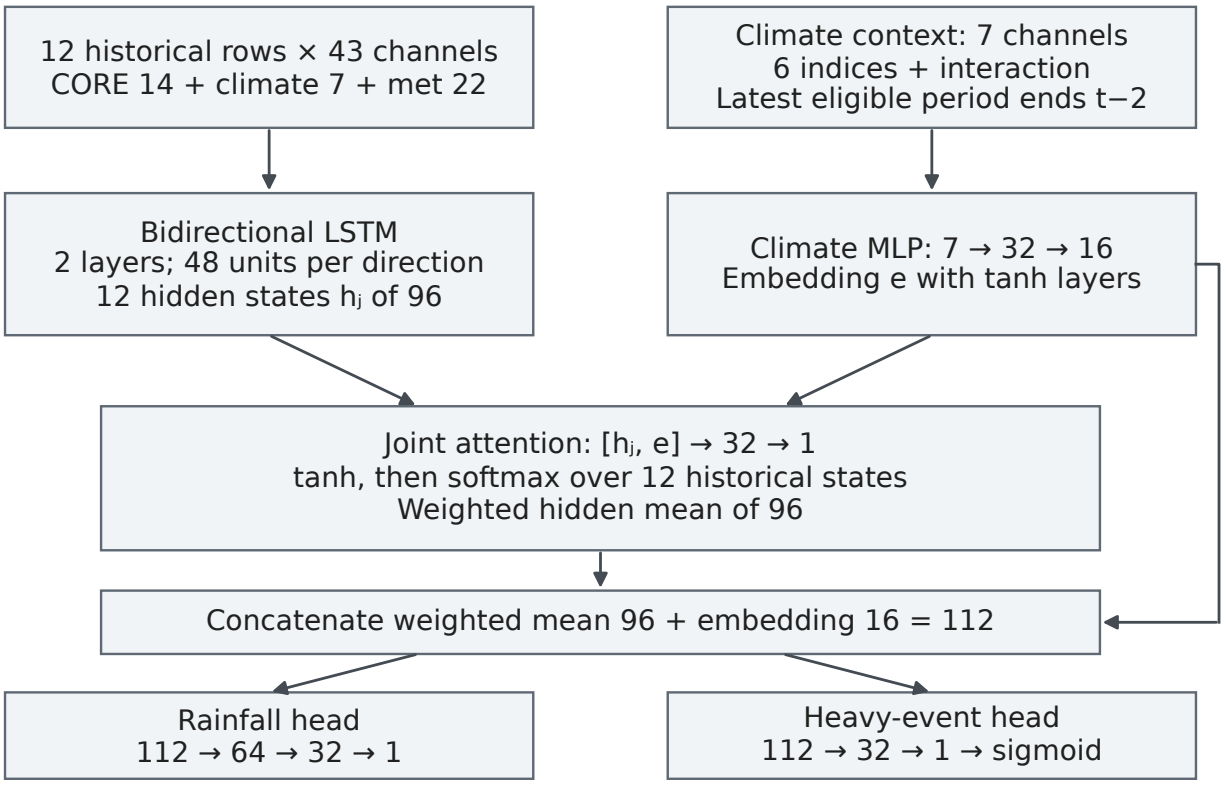


FULL ALCE architecture and the CAAA family training objective. Neural comparators share the two-layer bidirectional network; prespecified removal controls disable their designated components. Climate context enters nonlinear joint attention and is also concatenated directly with its weighted hidden mean. The heavy-event head produces logits during training with BCEWithLogitsLoss and probabilities through sigmoid for reporting. The event is observed rainfall strictly greater than each client's TRAIN q80; the fixed probability cutoff is 0.5. Inputs use shared TRAIN scaling; target scaling and q80 are fitted locally. CAAA climate relevance uses fixed whole-TRAIN severity and two local sensitivity summaries; inner-monitor performance varies by round. The proximal coefficient uses eligible batch-example severity. Outer validation selects hyperparameters and global/personal checkpoints, never gradient updates. The selected primary aggregation coefficients are (1,0,0), so its performance and climate aggregation terms are inactive; the anchor removal experiments use (0.3,0.3,0.4). This is a one-process simulation with plain summaries and no secure aggregation or differential privacy. Graph controls require separate privileged access to aligned TRAIN rainfall; centralized and static-router controls use declared raw pooling.